

Derivatives

Note that we can also define higher-order derivatives

$$f''(a) = \lim_{h \rightarrow 0} \frac{f'(x+h) - f'(x)}{h} \text{ etc.}$$

The existence of higher-order derivatives are impossible to detect by pictures.

Ex $f(x) = x^3$

$$f'(x) = 3x^2 \text{ differentiable}$$

$$g(x) = \begin{cases} x^2, & x \geq 0 \\ -x^2, & x < 0 \end{cases}$$

$$g'(x) = \begin{cases} 2x, & x \geq 0 \\ -2x, & x < 0 \end{cases} \text{ Not differentiable}$$

XI. The Significance of Derivatives

Def Let $f: D \rightarrow \mathbb{R}$, $x \in D$ is a maximum pt of f

$$\text{If } f(x) \geq f(y) \quad \forall y \in D$$

- Maximum may not exist

$$f(x) = \frac{1}{x}, \quad D = \mathbb{R} \setminus \{0\}.$$

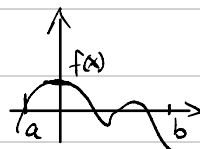
$$f(x) = x, \quad D = \{a, b\}$$

- If D is a compact ($\Rightarrow$ closed + bounded), then max exists by EVT.

Thm function f on (a, b) , $x \in (a, b)$ max pt. of f , f is diff. at x ,

$$\text{Then } f'(x) = 0$$

Pf WTS $\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = 0$



Idea If h very small $\Rightarrow f(x+h) \leq f(x)$

$$\text{If } h > 0, \begin{cases} f(x+h) - f(x) \leq 0 \\ h > 0 \end{cases} \Rightarrow f'(x) \leq 0$$

$$\text{If } h < 0, \begin{cases} f(x+h) - f(x) \leq 0 \\ h < 0 \end{cases} \Rightarrow f'(x) \geq 0$$

Since $f'(x)$ exists, $f'(x) = 0$

If you want to find a max pt. x

$$1^{\text{st}} \text{ derivative test } \left\{ \begin{array}{l} \cdot f'(x) = 0 \\ \cdot f \text{ not diff at } x \\ \cdot \text{ If } f: [a, b] \rightarrow \mathbb{R} \\ \text{the end pts are} \end{array} \right.$$